

INTELLIGENT SYSTEM DEVELOPMENT

Assignment 01

From Data Representation to a First Intelligent System

“Understand the system, represent the information, learn from data, and turn the model into an application.”

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Duration: 2 weeks

Due date:.....

Mode: Individual

1 Purpose of Assignment 01

Assignment 01 is the first practical step after Slide 01, *Intelligent System Development*. Slide 01 introduces the course-level progression:

Intelligence → Intelligent System → Representation → Learning → Decision → Action

and the development philosophy:

Understand → Represent → Implement → Experiment → Apply

Assignment 01 makes the first concrete realization of this framework using structured data and traditional machine learning.

Foundation

Central idea

A traditional ML system does not directly operate on the whole real world. It operates on a computational representation of selected information.

Real-world problem → Data → Feature representation → ML model → Prediction

The goal is therefore not merely to “train a classifier.” The goal is to develop and explain a small intelligent system.

2 Connection with Slide 01

Students should explicitly connect every part of Assignment 01 to the conceptual framework introduced in Slide 01.

Slide 01 concept	Assignment 01 realization	Student evidence
Intelligence	Predictive capability	Meaningful prediction task
Intelligent system	Data + representation + model + application	System diagram
External information	Dataset / observations	Dataset description
Internal computational representation	Feature vector	Feature table and X
Learning	Training a traditional ML model	Training procedure
Decision	Prediction $\hat{y}$	Model output
Action / application	Use prediction in a small application	Working demo
Development	Experiment, evaluate, improve	Controlled experiments

Important

The assignment must be understandable as a small **intelligent system**, not as a collection of unrelated Scikit-learn examples.

3 What Is the Intelligent System Being Developed?

Before writing code, students must define the system.

A useful abstraction is:

Environment $\rightarrow$ Input $\rightarrow$ Representation $\rightarrow$ Learning / Model $\rightarrow$ Decision $\rightarrow$ Output

For this assignment:

Real-world data $\rightarrow$ Feature vector $\rightarrow$ Traditional ML $\rightarrow$ Prediction

Students must answer:

1. What real-world problem does the system address?
2. What information does the system receive?
3. How is that information represented internally?
4. What does the model learn?
5. What decision or prediction does it produce?
6. Who or what uses the prediction?

Student Requirement

Required system statement

Write a short paragraph describing the intelligent system before starting model training. Also draw a simple system diagram showing:

Input $\rightarrow$ Representation $\rightarrow$ Model $\rightarrow$ Prediction.

4 Intelligence in This Assignment

In this first assignment, intelligence is deliberately treated in a limited and operational sense. The system demonstrates a form of intelligence when it can:

- receive information;
- represent relevant information computationally;
- learn a relationship from examples;
- produce a prediction for a new case;
- support a decision or application.

Thus:

Learning from examples + generalization + prediction

is the working notion of intelligence for Assignment 01.

This is not intended to be a complete definition of intelligence. Later assignments will extend the notion through deep learning, knowledge, retrieval, generation, interaction, and deployment.

5 Data Representation: The Central Idea

A real-world object or event can be represented in different ways. For structured data, an observation is represented as a feature vector:

$$x_i = [x_{i1}, x_{i2}, \dots, x_{id}] \in \mathbb{R}^d.$$

A supervised dataset is:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N.$$

Here:

- x_i is the input representation;
- d is the number of features;
- y_i is the target;
- N is the number of observations.

The complete feature matrix is:

$$X \in \mathbb{R}^{N \times d}.$$

Foundation

Key principle

The model does not receive “the real world.” It receives a representation of selected information. Therefore:

Representation determines what information is available to learning.

6 Why Traditional Machine Learning Fits Here

Slide 01 places traditional ML in the following pipeline:

Raw / Domain Data → Human-designed Features → Traditional ML Model → Prediction

This is the central technical focus of Assignment 01.
Students should understand the distinction between:

Traditional ML : human-designed feature representation + learned prediction model

Deep Learning : learned representation + learned prediction model.

The second approach will be studied later.

7 Dataset Selection

Students must select one real structured dataset from Kaggle or another approved public source.
The dataset should contain:

- a meaningful real-world problem;
- identifiable observations;
- usable features;
- a clearly defined target;
- enough data for a meaningful experiment;
- a problem suitable for traditional ML.

Possible domains include:

- healthcare;
- finance;
- education;
- agriculture;
- business;
- e-commerce;
- transportation;
- environment.

Students must [record the dataset source](#).

Student Requirement**Dataset questions**

Before training any model, answer:

1. What real-world phenomenon is represented?
2. What is one observation?
3. What are the features?
4. What is the target?
5. Is the target numerical or categorical?
6. Is this regression or classification?
7. How many observations are available?
8. How many features are available?
9. Which features are numerical?
10. Which features are categorical?

8 Data Representation Task

Students must explicitly construct the representation used by the model.

For example:

$$x_i = [\text{age}, \text{income}, \text{BMI}, \text{duration}, \dots].$$

Provide a table:

Feature	Type	Representation	Meaning
Age	Numerical	Real value	Age of subject
Gender	Categorical	Encoded value	Category
BMI	Numerical	Real value	Body mass index
Status	Categorical	Binary/encoded	Status

Explain how categorical variables are represented numerically when required.

Students must distinguish:

$$\boxed{\text{Raw feature} \neq \text{encoded feature} \neq \text{model input}}$$

when preprocessing changes the representation.

9 Formulating the Learning Problem

The supervised learning problem is:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N.$$

The model learns:

$$\hat{y} = f_{\theta}(x).$$

The learning objective can be expressed as:

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(x_i), y_i).$$

Students should be able to explain:

- input;
- target;
- model;
- parameters;
- prediction;
- loss/error;
- training;
- testing;
- generalization.

10 Regression or Classification?

Students must identify the learning problem before selecting models.

10.1 Regression

A regression system predicts a numerical quantity:

$$x \rightarrow \hat{y} \in \mathbb{R}.$$

Examples include:

- house price;
- sales amount;
- temperature;
- energy consumption.

10.2 Classification

A classification system predicts a class:

$$x \rightarrow \hat{y} \in \{1, \dots, K\}.$$

Examples include:

- spam / not spam;
- disease / healthy;
- churn / no churn;
- benign / malignant.

Important

The report must contain a one-sentence formal problem statement.
For example:

Given the selected feature vector, predict the target class for a previously unseen observation.

11 Train/Test Separation

Students must separate data used for learning from data used for evaluation.

A simple split is:

$$D = D_{\text{train}} \cup D_{\text{test}}.$$

For example:

80% training, 20% testing.

Example:

```
1 from sklearn.model_selection import train_test_split
2
3 X_train, X_test, y_train, y_test = train_test_split(
4     X,
5     y,
6     test_size=0.20,
7     random_state=42,
8     stratify=y
9 )
```

Students must explain why the test set must not be used as a training source.

12 Baseline: The First Reference Point

Before building sophisticated models, students must establish a baseline.

For classification:

```
1 from sklearn.dummy import DummyClassifier
2
3 baseline = DummyClassifier(
4     strategy="most_frequent"
5 )
6
7 baseline.fit(X_train, y_train)
8 y_pred = baseline.predict(X_test)
```

The student must answer:

Why is a baseline necessary?

The purpose is to determine whether the learned model provides meaningful improvement over a simple strategy.

13 Traditional ML Models

Students must train at least four traditional ML models. Recommended classification models:

1. Logistic Regression;
2. k-Nearest Neighbors;
3. Decision Tree;
4. Random Forest;
5. Support Vector Machine.

For regression, suitable choices include:

1. Linear Regression;
2. Decision Tree Regressor;
3. Random Forest Regressor;
4. Support Vector Regression.

Students do not need to implement the algorithms completely from scratch, but must understand their basic learning principles.

14 Model Understanding

For every selected model, explain:

1. What representation does it receive?
2. What relationship does it try to learn?
3. What parameters or structures are learned?
4. What criterion guides learning?
5. What assumptions does the model make?
6. What are its strengths?
7. What are its weaknesses?

Examples:

14.1 Linear Regression

$$\hat{y} = w^T x + b.$$

14.2 Logistic Regression

$$z = w^T x + b, \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

14.3 k-NN

Prediction is based on nearby training observations according to a chosen distance measure.

14.4 Decision Tree

The feature space is recursively divided using selected splitting criteria.

14.5 Random Forest

Multiple decision trees are combined:

many trees $\rightarrow$ combined prediction.

14.6 SVM

The model seeks a separating boundary with an appropriate margin.

15 Controlled Experiments

Students must conduct at least three controlled experiments.

15.1 Experiment Model Comparison

Compare at least four models under the same evaluation protocol.

Model	Accuracy	Precision	Recall	F1
Model 1	–	–	–	–
Model 2	–	–	–	–
Model 3	–	–	–	–
Model 4	–	–	–	–

15.2 Experiment Hyperparameter Investigation

Change one meaningful hyperparameter.

Examples:

- k in k-NN;
- tree depth;
- number of trees;
- SVM regularization;
- regularization in Logistic Regression.

The experimental question must be stated before running the experiment.

15.3 Experiment Representation / Feature Investigation

This experiment directly responds to Slide 01.

Compare, for example:

X_{all} vs. X_{selected} ,

or:

unscaled features vs. standardized features.

Students must explain whether the representation change affects the result and why.

16 Evaluation

For classification, report:

- Accuracy;
- Precision;
- Recall;
- F1-score;
- Confusion Matrix.

Important definitions are:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN},$$

$$Precision = \frac{TP}{TP + FP},$$

$$Recall = \frac{TP}{TP + FN},$$

$$F1 = \frac{2PR}{P + R}.$$

For regression, report:

- MAE;
- MSE;
- RMSE;
- R^2 .

Students must explain which metrics are appropriate for their application.

17 From Model to Intelligent Application

A model alone is not the complete system.

Assignment 01 therefore requires students to turn the trained model into a small application.

The application should realize:

Input → Representation → Preprocessing → Model → Prediction → Output

The same representation used during training must be used for new inputs.

Example:

```

1 def predict(model, sample):
2     # Convert application input into
3     # the same representation used in training.
4     prediction = model.predict(sample)
5     return prediction

```

The student must demonstrate at least three input cases.

Application

System requirement

The final demonstration must show the complete path:

User / Environment $\rightarrow$ Input $\rightarrow$ Feature Representation $\rightarrow$ ML Model $\rightarrow$ Prediction

This is the first small intelligent system developed in the course.

18 Reflection: What Makes This an Intelligent System?

This section is mandatory.

Students must answer:

1. What information does your system receive?
2. What is the internal representation?
3. What does the model learn from examples?
4. What prediction or decision does it make?
5. Why can it handle an unseen input?
6. What part of the system can reasonably be called “intelligent”?
7. What limitations prevent it from being a more capable intelligent system?

Students should distinguish:

trained model $\neq$ complete intelligent system

A complete system also requires input handling, representation, preprocessing, prediction, output, and integration into an application.

19 Reflection on Representation

Students must answer:

1. Why is your feature-vector representation appropriate?
2. What information does it preserve?
3. What information might it lose?
4. Could the same problem be represented as an image?
5. Could it be represented as a sequence?
6. Could it be represented as a graph?
7. Could it be represented using learned embeddings?
8. What would change if the representation changed?

The purpose is to prepare students for the central course idea:

Different information representations $\Rightarrow$ different learning methods

20 Connection to the History of AI

Slide 01 presents intelligent-system development as an evolving history of representations and learning mechanisms.

Assignment 01 occupies one important stage:

Structured Data → Feature Engineering → Traditional ML

Students should briefly explain where their system belongs in the broader evolution:

Symbolic AI	→	Explicit rules and symbols
Statistical ML	→	Features + learned models
Deep Learning	→	Learned representations
Foundation / Generative AI	→	Large learned representations + generation
Interactive / Agentic Systems	→	Perception + reasoning + action + interaction.

The purpose is not to study the complete history in Assignment 01. The purpose is to understand that traditional ML is one stage in the development of intelligent systems.

21 Required Notebook Structure

The Jupyter Notebook must contain:

1. System and Problem Definition
2. Intelligent System Diagram
3. Dataset Source
4. Dataset Description
5. Data Representation
6. Feature and Target Analysis
7. Exploratory Data Analysis
8. Train/Test Split
9. Baseline
10. Model 1
11. Model 2
12. Model 3
13. Model 4
14. Evaluation
15. Experiment 1: Model Comparison
16. Experiment 2: Hyperparameter Investigation
17. Experiment 3: Representation / Feature Investigation
18. Final Model
19. Application

20. System Demonstration
21. Reflection
22. Conclusion

22 Explicit Requirements

Student Requirement

R1. Real Dataset

Use a real dataset from Kaggle or another approved public source.

R2. System Definition

Define the real-world problem and the intended intelligent system.

R3. Representation

Explicitly describe how raw information becomes model input.

R4. Feature Analysis

Explain the meaning and type of important features.

R5. Problem Formulation

Clearly define input, target, and learning task.

R6. Baseline

Implement and report a baseline.

R7. Multiple Models

Train at least four traditional ML models.

R8. Controlled Experiments

Perform at least three experiments.

R9. Evaluation

Use appropriate metrics and explain them.

R10. Scientific Analysis

Explain why the results were obtained.

R11. Application

Develop a small application using the selected model.

R12. System Demonstration

Demonstrate the complete input-to-prediction pipeline.

R13. Reflection

Explain the role and limitations of the representation.

R14. Reproducibility

The notebook must run from beginning to end with documented environment and dataset information.

23 What Students Must NOT Do

The following are not sufficient:

- Downloading a dataset and immediately calling `model.fit()`.
- Reporting only accuracy.
- Training only one model.
- Copying code without understanding it.
- Changing many parameters without an experimental question.
- Claiming that a model is “better” without evidence.

- Using the test set repeatedly during model development.
- Presenting results without explaining them.
- Showing a trained model without demonstrating the complete application pipeline.

Important

The assignment evaluates **understanding, representation, experimentation, and system development**, not the amount of code.

24 Deliverables

Each student/team must submit:

1. Jupyter Notebook (.ipynb);
2. dataset information and source;
3. technical report;
4. system architecture / pipeline diagram;
5. experimental comparison table;
6. figures and visualizations;
7. application source code;
8. README describing how to run the system.

25 Suggested Report Structure

The report should contain:

1. Introduction
2. Intelligent System Definition
3. Problem Definition
4. Dataset
5. Data Representation
6. Traditional ML Methods
7. Experimental Design
8. Results
9. Model Comparison
10. Representation Analysis
11. Intelligent Application
12. Limitations
13. Reflection
14. Conclusion

26 Assessment Rubric

Component	Weight
Intelligent-system and problem understanding	15%
Data and representation analysis	15%
ML model understanding and implementation	15%
Experimental design	15%
Evaluation and model comparison	15%
Scientific analysis and reflection	15%
Application and system demonstration	7%
Code/report quality and reproducibility	3%
Total	100%

27 AI-Assisted Development

Students may use AI-based programming assistants such as ChatGPT during development.

However, the student remains responsible for the submitted work.

Students must:

- understand generated code;
- verify generated code;
- understand all reported results;
- explain the selected algorithms;
- explain the experimental conclusions;
- cite external sources when appropriate;
- disclose substantial AI assistance when required by course policy.

During demonstration, students may be asked to explain any algorithm, code component, experiment, or result.

28 Connection to Assignment 02 and the Course Roadmap

Assignment 01 establishes the first concrete layer:

Structured Data → Feature Vector → Traditional ML → Prediction
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The course then progressively expands the representation and intelligence capabilities:

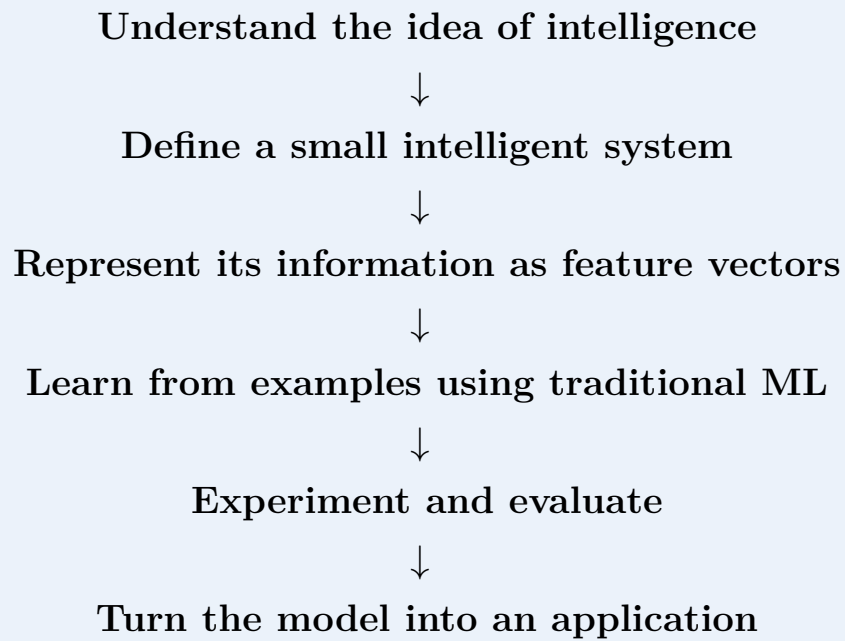
- | | |
|----------------|--|
| A1 | Feature vectors + Traditional ML |
| A2 | Data preparation + evaluation + better representations |
| A3 | Learned representations + Deep Learning |
| A4 | Tensors + CNN + structured knowledge / graphs |
| A5 | Embeddings + RAG + interaction + deployment |
| Project | Integrated Intelligent System |

The long-term course progression is:

Data → Representation → Learning → Knowledge → Retrieval → Generation → Interaction → Intelligent System
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29 Final Assignment Message

From Slide 01 to Assignment 01



Understand → Represent → Learn → Experiment → Apply

GOOD LUCK!